

SISIMIUT HOLSTEINSBORG, Greenland

WMO: 042340

Lat: 66.9510N

Lon: 53.7200W

Elev: 26

StdP: 14.68

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.7	-10.9	-24.1	1.4	-13.1	-19.9	1.8	-8.2	38.4	35.5	33.2	29.3	4.1	100	0.882

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	61.6	49.1	57.6	48.0	54.0	46.6	50.8	58.0	49.2	55.5	47.5	53.0	14.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
47.6	48.9	51.1	45.5	45.1	49.9	43.9	42.4	48.5	20.7	58.3	19.8	55.6	19.0	53.2	58.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	24.1	20.7	-15.2	67.6	7.6	4.6	-20.7	70.9	-25.1	73.6	-29.4	76.2	-34.9	79.5		
(5)			-15.9	53.9	7.2	2.9	-21.1	56.0	-25.3	57.6	-29.3	59.2	-34.6	61.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	12.1	9.4	11.0	23.1	33.4	41.9	46.6	45.7	39.4	30.5	23.7	18.0		
	DBStd	15.43	11.11	12.77	12.25	8.84	6.16	5.46	5.44	4.61	4.28	6.09	7.60	8.68		
	HDD50	8084	1176	1138	1209	806	515	251	135	148	318	606	789	993		
	HDD65	13507	1641	1558	1674	1256	980	694	572	597	768	1070	1239	1458		
	CDD50	51	0	0	0	0	0	6	28	16	1	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	7.8	9.1	7.6	7.5	7.5	7.5	7.6	7.2	7.0	7.6	7.5	8.6	9.0		
	PrecAvg	23.2	2.2	1.4	1.4	1.4	1.3	1.4	1.9	2.6	2.6	2.2	2.5	2.4		
(15) Precipitation	PrecMax	32.4	6.2	3.4	3.6	3.2	2.3	3.0	3.6	4.3	5.6	4.6	5.3	5.0		
	PrecMin	15.4	0.2	0.6	0.1	0.5	0.3	0.5	0.7	0.9	0.7	0.6	0.9	1.0		
	PrecStd	4.2	1.4	0.7	0.8	0.7	0.6	0.6	0.8	0.9	1.0	0.9	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	44.2	43.2	44.9	55.6	63.8	68.2	64.4	54.3	49.5	47.0	42.7		
		MCWB	34.5	34.5	36.0	38.1	44.2	48.8	51.9	51.0	45.7	40.0	39.3	34.8		
	2%	DB	37.0	38.2	37.3	39.8	48.0	58.2	62.9	59.3	50.0	44.0	41.0	37.8		
		MCWB	31.9	32.2	32.7	34.7	41.1	46.4	50.0	49.5	42.8	37.3	35.1	33.6		
	5%	DB	32.0	32.6	33.1	37.1	44.3	53.5	58.4	55.4	47.3	40.8	37.2	34.0		
		MCWB	28.3	28.6	29.4	33.3	38.5	44.9	48.4	48.0	41.3	36.1	33.3	30.4		
	10%	DB	27.5	26.7	28.7	34.7	41.2	50.0	54.9	52.2	45.3	38.2	34.3	29.8		
		MCWB	24.3	23.8	25.6	31.7	36.4	43.3	47.3	46.5	40.4	34.2	31.2	26.7		
	0.4%	WB	37.1	35.9	37.0	38.8	44.9	50.3	54.0	53.0	48.3	41.9	41.3	36.8		
		MCDB	40.6	41.3	42.4	43.7	52.6	60.5	62.7	61.0	51.8	45.9	45.9	40.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.0	32.3	32.9	35.6	41.8	47.4	51.2	50.7	44.1	38.6	36.4	33.9		
		MCDB	35.3	38.1	36.5	38.6	47.3	55.6	59.2	56.5	47.8	42.5	39.2	37.6		
	5%	WB	28.5	28.6	29.6	33.8	39.1	45.5	49.5	49.1	42.6	36.6	33.6	30.2		
		MCDB	32.2	33.1	33.0	36.3	43.1	52.2	56.7	54.1	45.9	40.0	36.8	33.6		
	10%	WB	24.4	23.6	25.6	31.9	37.0	43.7	47.8	47.5	41.1	34.6	31.3	26.7		
		MCDB	27.5	26.7	28.8	34.7	40.3	49.3	53.9	51.6	44.4	37.8	34.3	29.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.9	10.6	10.0	9.1	11.2	11.5	9.3	7.5	7.4	7.8	8.0		
		MCDBR	14.2	16.3	13.9	11.8	15.5	19.5	19.7	16.2	10.7	10.0	11.5	11.1		
		MCWBR	11.1	12.6	10.4	8.2	9.1	9.9	9.2	8.1	6.4	7.3	9.2	8.5		
	5% WB	MCDBR	13.9	16.3	13.8	10.6	14.6	17.6	17.4	14.5	9.1	8.9	10.9	10.9		
		MCWBR	11.2	13.1	10.8	7.8	9.0	9.4	8.9	7.9	5.9	7.1	9.3	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.170	0.222	0.255	0.313	0.320	0.332	0.333	0.334	0.293	0.248	0.356	N/A		
	taud		1.929	2.213	2.316	2.239	2.381	2.476	2.520	2.528	2.540	2.260	2.850	N/A		
	Ebn at Noon		69	208	261	267	279	278	274	262	248	189	77	0		
	Edh at Noon		4	14	23	32	31	30	27	24	19	13	5	0		
(44) All-Sky Solar Radiation	RadAvg		16	162	634	1242	1714	1848	1674	1155	667	246	37	0		
	RadStd		2	20	52	107	101	164	161	89	66	21	5	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+2.14	N/A	N/A	+3.85	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+1.64	N/A	N/A	+1.48	N/A	N/A	N/A	N/A	+14	N/A

Nomenclature: See separate page